Module - CE4023 HYDRAULIC DESIGN

Assignment	**Field Exercise on Hydraulic Structures (LO1)**	Marks	10 %

		Students Response
Learning Outcomes of the Practical	• *Evaluate* the suitability of hydraulic structures for intended uses/ applications.	Yes/No
	• *Determine* key hydro-meteorological parameters required for hydrological designs.	Yes/No
	• *Assess* the reservoir capacity, yield, and spillway capacity.	Yes/No
	• *Design* hydraulic structures by considering hydrologic, hydraulic, economic, and environmental factors.	Yes/No

Programme Outcomes	PO1: Apply knowledge of mathematics, basic sciences and engineering fundamentals to solve complex engineering problems	H
	PO2: Identify, formulate, research literature, and analyze complex engineering problems to arrive at valid conclusions.	M
	PO3: Design solutions to complex engineering problems; involving systems, components or processes that meet specified needs, including conflicting constraints.	H
	PO4: Conduct investigations of complex engineering problems using research based knowledge and research methods (including experimental design, analysis & interpretation of data and synthesis of information)	H
	PO5: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools to complex engineering activities	M
	PO6: Assess societal, health, safety, legal, cultural and environmental issues related to professional engineering solutions	M
	PO7: Demonstrate broad knowledge of sustainable development concepts and practices required for dealing with contemporary issues related to professional engineering practice.	M
	PO8: Demonstrate broad knowledge of ethical responsibilities and professional standards	
	PO9: Demonstrate ability to function effectively as an individual and in multidisciplinary and multi-cultural teams, with the capacity to be a leader or manager as well as an effective team member.	
	PO10: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	M
	PO11: Demonstrate broad knowledge of management and business practices, including financial management, risk and change management.	L
	PO12: Engage in independent and lifelong learning in the broad context of technological change.	M
H – High	M – Medium L– Low	
Lecturer	Prof. R. L. H. L. Rajapakse, Dr. H. G. L. N. Gunawardhana	

Student Name:			
Registration Number:			
Date of Assignment:		**Date Due:**	
Initial Submission Date:		**Re Submission Date:**	

Important

1. Please note that **plagiarism** is treated as a serious offence and therefore the work you produce **must be individual and original.**
2. All sources of information must be **referenced using "Harvard Referencing**" where a **reference list / Bibliography** should be included at the end of the assignment. (You may refer the information given in **http://libweb.anglia.ac.uk/referencing/harvard.htm**)
3. Please note that the submission date given for the assignment (14 days after the date of performance for the laboratory session) is the **final date** that you can submit the assignment. If the given submission date is a public holiday, redefined submission deadline will be at 1300 hr of the immediate following working day. Late submissions **will be graded lower.**
4. Assignments returned to students for corrections must be **re-submitted within 10 days.**
5. **Failure to re-submit** the previously marked assignment with the re-submitted assignment will mean that **results cannot be released for the respective unit**.

Plagiarism

Although research and discussion form an essential part of the assignment, deliberate copying of the work of others or unacknowledged copying from printed or electronic sources is NOT permitted. Disciplinary actions will be taken against those who are found guilty of plagiarism. Signing of this sheet is required to indicate your compliance with the above regulations.

Student's Signature: …….. Date: ……

Student's Comments (Prior to the submission) if any:

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Student's feedback (After the submission) if any:

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Identification of general hazards associated with the practical

Hazardous Chemicals, Substances, Biohazards

Chemicals/Substances

- ☐ Compressed Gases - Flammable
- ☐ Compressed Gases- Oxidizing
- ☐ Compressed Gases- Toxic
- ☐ Compressed Gases - Inert
- ☐ Cryogenic materials
- ☐ Organic peroxides
- ☐ Peroxide Formers
- ☐ Self-reactive substances
- ☐ Water-reactive substances
- ☐ Pyrophorics
- ☐ Acutely toxic chemicals
- ☐ Carcinogens
- ☐ Nanomaterials
- ☐ Reproductive Toxins
- ☐ Simple Asphyxiant
- ☐ Corrosive Liquid
- ☐ DEA/Controlled Substances
- ☐ Specific Organ Toxicity
- ☐ Explosives
- ☐ Flammable Liquids
- ☐ Oxidizers/reducing agents

Biohazards

- ☐ Animal Infection Studies
- ☐ Large Scale Culture
- ☐ Risk Group – 2 Pathogens
- ☐ Risk Group – 3 Pathogens
- ☐ Plant Pathogens
- ☐ Biological Toxins
- ☐ Human, Blood, Body Fluids
- ☐ Cell Culture
- ☐ Viruses
- ☐ Transgenic Plants or Animals

Hazardous Processes or Equipment

- ☐ Explosion hazard
- ☐ Exothermic, with potential for fire or excessive heat
- ☐ Acid Baths
- ☐ Hazardous reaction or products
- ☐ Generation of air contaminants (e.g. gases, aerosols, particulates)
- ☐ Heating chemicals
- ☐ Large volumes
- ☐ Chemical transferring
- ☐ Hand/power tools
- ■ Moving Equipment or parts
- ■ Electrical hazards
- ☐ Noise > 85 dBa
- ☐ Hot surfaces
- ☐ Ergonomic Hazard
- ☐ Needles/Sharps
- ☐ Drying Oven/Furnace
- ☐ Centrifuge
- ☐ Working alone/ Afterhours
- ☐ Unattended Reactions
- ☐ Respiratory Hazard
- ☐ Vacuum/Pressure Systems
- ☐ Refrigerators and Freezers
- ☐ Stirring and Mixing devices
- ☐ Laboratory Microwave ovens
- ■ Slip, trip, falls
- ☐ Repetitive Motion/vibrations

Field Hazards

- ☐ **Foul Weather**
- ☐ **Temperature Extremes**
- ☐ **Darkness/low light**
- ☐ **Altitude**
- ☐ **Smoke/dust/mist**
- ☐ **Wild Animals/insects**
- ☐ **Plants/Allergens**
- ☐ Vector-borne or other endemic disease
- ☐ Hygiene/water or food-borne illness
- ☐ **Falling objects**
- ☐ **Boating/swimming/water hazards**
- ☐ Limited Communication
- ☐ **Remote area/limited medical services**
- ☐ Lifting/carrying
- ☐ Strenuous physical activity (e.g. long days, high stress, etc.)
- ☐ Driving/Operating a Vehicle (e.g. Tractor, OSU or personal)
- ☐ **Uneven Surfaces**
- ☐ **Heights**

Shop/Laser/Radiation

Shop

- ☐ Aerial Lift
- ☐ Air Compressor
- ☐ Crane
- ☐ Forklift
- ☐ Hot Work
- ☐ Used/New Oil

Laser

- ☐ Class IIIb Laser
- ☐ Class IV Laser

Radiation

- ☐ X-ray Machine
- ☐ Magnetic Field (e.g. NMR, MRI)
- ☐ Radioactive Materials
- ☐ Unsealed Source Radionuclides
- ☐ Sealed Source Radionuclides
- ☐ Ultraviolet Light/Infared Light

Required Laboratory PPE/Safety Supplies

- ■ Appropriate clothing (long pants, closed-toed shoes, etc)
- ☐ Gloves; indicate type: _____
- ☐ Safety glasses
- ☐ Safety goggles
- ☐ Hearing protection
- ☐ Face shield and goggles
- ☐ Hard hat
- ☐ Lab coat
- ☐ Flame-resistant lab coat
- ☐ Fire extinguisher
- ☐ Eyewash/safety shower
- ☐ First aid kit
- ☐ Spill kit
- ☐ Specialized medical supplies (e.g., calcium gluconate for hydrofluoric acid)
- ☐ Others (list):……

Risk Assessment Tool

Risk Matrix		Consequence/Severity		
		Low	Medium	High
Likelihood/ Probability	High (almost certain)	*Medium*	*High*	*High*
	Med (likely/possible)	*Low/Medium*	*Medium*	*High*
	Low (rare)	*Low*	*Low*	*Medium*

The following table defines what a low, medium and high severity risk is based on risk to people, environment, operations, and reputation. Use this table to assist with the Risk Matrix above.

Consequence/ Impact					
		People	Environment	Operations	Reputation
Severity	Low	No safety or health hazards present, first aid injury, short-term exposure to mild health effect/field accidents.	Minor weather or field/ Slight effect.	Insignificant or minimal impact on operations.	Lab/ Department/ level attention.
	Medium	Medical treatment injury, chronic health effects, time-loss <7 days, moderate health effect.	Controlled environmental release, minor and/or localized effects.	Could delay operations due to minor property damage.	Local area attention, University level attention.
	High	Severe health effect, time-loss > 7 days, permanent incapacitation or disability of fatality.	Some permanent /widespread ecological damage, major effect	Significant or long-term effects on the ability to continue operations, results in significant property damage and financial loss.	State, National and International attention.

The following table defines what a low, medium and high likelihood/probability is for the Risk Matrix tool.

Likelihood/Probability	Explanation
Low	Nearly unlikely to happen in the near future and no immediate action is needed.
Medium	Moderately unlikely or likely to occur and actions should be taken to reduce or control the risk.
High	More than likely to occur or high probability the risk will occur; immediate action plans required.

The risk rating for the practical w.r.t. the above assessment tool:	Low

Student feedback on laboratory health and safety practices

Are satisfactory post-experiment processes adapted? (e.g., waste disposal, material reuse, cleaning equipment, floor/tables, switch off unused equipment, etc.)	**Yes/No**
If not, please suggest the post-experimental measures you propose.	
Are you satisfied with the overall occupational safety of the laboratory? (e.g., safety during the experiment, adequate lighting and ventilation, unobstructed exits, availability and easy access to first aid box, fire extinguishers, availability of trained staff in an emergency etc.)	**Yes/No**
If not, what additional safety measures do you propose to improve the overall safety of the laboratory and/or the safety during the experiment?	

I am aware of the general health and safety practices adapted in the laboratory and have read and understood the general hazards and risks associated with the practical.

Student's Signature: …….. **Date:……**

Hydraulic Structures Field Study Exercise

CE4023: Hydraulic Design

Hydraulic Engineering Laboratory
Department of Civil Engineering
University Of Moratuwa
Sri Lanka

ASSIGNMENT:

Each student needs to visit* at least two locations (at least 3 kilometers apart), where they have closely observed Hydraulic (water) Structures (at least one each at each location). One structure should be a very common structure and the other should not. **The structures should deal with freshwater.**

For Each Structure:

1) Take photographs (digital/clear resolution at postcard size) of the structure: Overall structure and key components related to Hydraulics (Water) and make sketches with approximate dimensions (elevations, plan, 3D view, sections, etc.)
2) Proof that you have visited the structure (a photograph with you and the structure and recognizable surrounding features)
3) Proof that you have visited the structure after the date mentioned by the lecturer.
4) Provide Coordinates of the Structure
5) Provide a Google extract of approximately 1:50,000 or a digital 1: 50,000 topo-sheet extract
6) Describe the Hydraulic engineering purpose of the structure
7) How are the aspects of (i) safety of the structure and (ii) safety of the environment looked after when you observe from the point of view of hydraulics?
8) What are the quantity and the head of water that (in your opinion) has been used for the design of the structure?
9) Provide justification for your opinion above
10) Submit the Hydraulic structure report (both hard copy and digital)
11) Download and show a picture of a similar structure from an internet reference and provide the URL with the date and time of visit to that site.

*Note: All students are strictly advised to follow and adhere to all health and safety guidelines, especially those related to Dengue/Viral flu/Rat Fever (Leptospirosis), etc., as well as, precautions against slips, trips and falls. If it is unsafe to visit structure locations physically (due to specific health or other constraints), you are allowed to use secondary data, online resources, etc. (with prior approval and adequate justification), to complete this assignment (prior approval should be obtained).

-- END OF ASSIGNMENT --

CE4023 Hydraulic Design - Hydraulic Structures Field Study Exercise

- Additional Notes

Assignment or the Problem

Visit at least two locations (at least 3 kilometers apart), and closely observe Hydraulic (water) Structures (at least one each at each location) which are to be associated with freshwater. One structure should be a very common structure and the other should not be a common structure type. Submit a "Hydraulic Structure Field Identification" report providing at least the following.

Provide a detailed map of the location. Provide the coordinates of each location. Photographs showing the entire structure and indicating important components such as inlets, conveyance section, energy dissipater, and outlet. Describe the type and purpose (hydraulic functionality) of the structures. Describe how the following aspects are taken care of (i) sustainability of the structure and (ii) sustainability of the surrounding environment

With appropriate justifications, make an estimate of the safe limits of the quantity and the head of water for each observed structure. Compare the observed structures with those of literature (you may use internet sources) and comment on the differences.

The report should be submitted both in digital and hardcopy format (digital format preferred).

ASSIGNMENT: FIELD IDENTIFICATION OF HYDRAULIC STRUCTURES THEIR PURPOSE AND PERFORMANCE

Identification of Hydraulic Structure Components, Purpose, and Functionality in relation to Hydraulic Design Module - A Visit to a Village Reservoir Site to Submit a Field Inspection Report

1. **Key Words**: The Hydraulic Design Key Words that You are expected to learn

   Earth Dam, Spillway, Sluice, Inlet, Outlet, Inlet and Outlet Transition, Stilling Basin, Cutoff Wall, Tail Canal, Flux, Free Board, Return Period, Measurement Device, Dead Storage, Minimum Operating Level, Main Drainage

2. **Methodology**: Identification of Hydraulic Structures and their Purpose

a. Visit a village reservoir which is managed by the Agrarian Services Department or by the Village Communities and NOT a large or a medium reservoir managed by another state agency such as Mahaweli Authority or Irrigation Department. (Capture the Coordinates of the Location - approximately center of the Bund)

b. Identify the Person Controlling the Water Issues (Name, Address and Contact for Verification, Take a picture of the person and You at the tank - Evidence of Execution)

c. Observe the Tank Bund, Spillway(s), Sluice(s), Take Photo(s) of the structures

d. Take notes of each structure (Typical Requirements are listed in the next section)

3. Reporting Requirements

a. The reporting and discussion should concentrate on Adequacy of the structure (Sustainability and Safety) from a strong engineering perspective

b. General Details of the Reservoir (Location, Name, Map, How many direct users, What are the main uses and what quantities; (ie, Quantity can be volume, persons, acres, seasons, or a combination based on the meaningfulness))

c. Objectives and Specific Objectives of the Assignment (purpose of carrying out the assignment)

d. Characteristics of the Spillway(s) (Condition report with a location sketch in a plan view of reservoir bund, Type of spill, length of spill, Drop in elevation when spilling, Spill and Stilling basin sketch, Length of stilling basin, drop of water up to stilling basin during spilling, downstream protection from erosion, cutoff walls, any other engineering aspect etc.,)

e. Characteristics of Spilling (On average how many time a year does it spill, What is the average flux and duration, in which months does this occur, What was the last occasion the

reservoir spilled, Any stakeholder comments on spilling, any other engineering aspect to judge the adequacy and safety of the structure)

f. Characteristics of the Sluice (s) (Condition report for each sluice with a location sketch in a plan view of reservoir bund, Type of sluice, sluice diameter or conveyance section dimension, Inlet, Outlet, Inlet and Outlet transitions, Gate Type and Condition, Trash racks, Approach path for Operation, measuring device, calibration, Stilling basin, Seepage cutoffs, sluice elevation)

g. Characteristic of Water Releases (from a stakeholder perspective, prevailing method of water control (water issues) and measurements, the waterway conditions and maintenance after the release, adequacy of dead storage and number of dry months)

h. Characteristics of the Reservoir Bund (Condition report, Bund Top width, Upstream Slope, Down stream slope, Seepage from Bund, Cracks on Bund, Is the bund top horizontal, the elevation differences between Full Supply Level (FSL), High Flood Level (HFL) and the Bund Top Level (BTL), what is the free board (FB))

i. Stakeholder concerns with respect to the design of the bund and associated structures (capacity adequacy, maintenance of the bund and structures, stakeholder role and division of work, wildlife issues, vegetation issues sharing of work, water theft etc.)

4. Discussion

a. What did you learn from this assignment? Mention issues faced by your and how they were overcome. Make recommendations for improvements.

b. If you had noticed any seepage from bund or cracks in bund, what are the reasons and remedial measures according to your engineering evaluation?

c. In your opinion what should be the dead storage for environmental sustainability for the driest months?

d. What should be the design return period for the spillway and bund? Indicate reasons for your judgment.

e. Indicate the important things that an engineer should consider when designing the sluice, bund and the turnouts

Note: Report should have your Name, Index Number, Date, and your passport size or similar photograph in the cover page; Include a copy of assignment instructions in the report. The report should be submitted in both hard copy and digital formats.

Submit your photographs for our teaching material - Save with your index and the name of District Visited.

-- END OF NOTE --